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Addressing Global Water Challenges in 2025: an Integrated Framework for Research, Policy, and Resource Management

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Abstract
Global water systems are under intensifying stress from climate variability, population growth, institutional fragmentation, and deteriorating infrastructure. This paper presents a structured framework addressing twenty of the most pressing water management challenges worldwide, categorized across four domains: irrigation, water supply, water quality, and water governance. For each challenge, it identifies key management issues, policy and research priorities, and regional relevance. Drawing on over 200 peer-reviewed sources as well as original hydro-economic modeling from North America's Rio Grande Basin, the paper illustrates how institutional innovation—particularly interdistrict and intradistrict water trading—can significantly enhance economic efficiency and adaptive capacity under drought. The model results highlight the superior performance of water markets relative to proportional shortage-sharing rules, demonstrating considerable gains in farm income and resource-use efficiency. More broadly, this paper's original contribution provides decision-support guidance for researchers, practitioners, and policymakers working at the intersection of hydrology, infrastructure, and governance. By integrating economic analysis with cross-sectoral and regional insights, the paper advances a practical roadmap for building more resilient, equitable, and sustainable water systems in an era of growing uncertainty. Addressing Global Water Challenges in 2025: An Integrated Framework for Research, Policy, and Resource Management.

Keywords Water governance · Climate adaptation · Water security · Sustainability · Research priorities · Hydrology · Infrastructure

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1 Introduction

1.1 Background

Water is fundamental to human well-being, agricultural productivity, public health, and economic development. Yet in 2025, billions of people still face unreliable or unsafe access to clean water. Water stress is intensifying internationally, driven by population growth, economic expansion, shifting consumption patterns, climate change, and deteriorating water quality. These pressures are compounded by weak institutions, fragmented governance, and chronic underinvestment in infrastructure. Many communities lack the technical, financial, and legal capacity to manage water risks effectively. As water becomes scarcer, competition intensifies across sectors, regions, and users—raising the stakes for reform.

1.2 Previous Work and Gaps

Much work has explored dimensions of water resource challenges, including groundwater depletion, infrastructure aging, irrigation inefficiency, and water pollution. Leading journals have published sectoral reviews focusing on hydrology (e.g., snowpack variability, evapotranspiration, stormwater runoff), water quality (e.g., nutrient loading, emerging contaminants), or governance failures (e.g., fragmented institutions, transboundary disputes). Several works also examine climate adaptation strategies, water pricing, trading mechanisms, or integrated water resource management (IWRM) at the basin scale.

Some of these works have been published in *Water Resources Management* (Villholth and Rajasooriyar 2010; Simonovic 2017; Warmink et al. 2017; Sondermann and de Oliveira 2021; Hasan et al. 2023) as well as more recent works (Devkota et al. 2025; Febriana et al. 2025; Li et al. 2025; Palazzolo et al. 2025; Yegin et al. 2025; Zhang et al. 2025). A number of other scientific journals have published important works summarizing global water resource management challenges (Kumar et al. 2006; Rockström et al. 2007; Roy et al. 2008; Wagener et al. 2010; Rothausen and Conway 2011; Mancosu et al. 2015; Van Loon et al. 2016; Chakona and Shackleton 2017; Gleeson and Richter 2018; Le and Nguyen 2024; Quoc 2024; Rattanarat et al. 2024).

Despite this progress, significant gaps remain: no work provides a systematic and integrated framework that synthesizes global water challenges across all four critical domains—irrigation, supply, quality, and governance—while identifying both research needs and actionable management strategies. Existing reviews tend to be fragmented—either disciplinary (e.g., hydrology or economics), regional (e.g., South Asia or Western U.S.), or sectoral (e.g., urban water, sanitation, or agriculture). As a result, decision-makers lack a coherent analytical roadmap to guide integrated, evidence-based water resource planning in the face of escalating climate, population, and institutional stress.

1.3 Original Contribution

This review article offers an original contribution to the field of water resources management. First, it presents a structured, cross-cutting synthesis of twenty critical global water challenges for which the selection is based on the criteria described in Table 1. These twenty

Table 1 Selection criteria for identifying global water management challenges

Criterion	Description
Global Relevance	The challenge affects multiple world regions and populations, ensuring it is not a localized or niche issue.
Scale of Impact	The problem has far-reaching consequences for human populations, economic systems, and ecological health.
Urgency and Escalation	The issue is becoming more severe over time due to compounding pressures such as climate change, population growth, or infrastructure decay.
Structural Nature	The problem reflects a root cause in water systems, rather than being a secondary or symptomatic issue.
Cross-Sectoral and Cross-Scale Complexity	The challenge spans sectors (e.g., agriculture, health, urban planning) and scales (local to international), requiring integrated management.
Representation of Diverse Water Domains	The set of challenges reflects a comprehensive view of water issues, including quantity, quality, governance, infrastructure, and ecosystems.
Recognition by Scholarly and Policy Communities	The challenge is widely recognized in academic literature and international frameworks such as the SDGs and UN Water reports.

issues were selected based on their frequency in literature, policy relevance, and regional diversity. The synthesis is organized into four interdependent domains: (1) irrigation, (2) water supply and reliability, (3) water quality, and (4) water governance capacity. Each challenge is systematically analyzed in terms of its problem characterization, research priorities, and geographic extent—producing a comparative, globally relevant management framework that has not previously been developed in the literature.

Second, the paper includes a new hydro-economic optimization model of water trading institutions in New Mexico's Rio Grande Basin, using positive mathematical programming to simulate inter- and intradistrict trading under drought. Third, by integrating synthesized literature with original modeling, the paper provides both a panoramic view of global water management gaps and a practical decision-support framework for adaptive planning under uncertainty. It explicitly links scientific insight to actionable strategies, emphasizing tradeoffs, governance levers, and research frontiers. As such, it serves the dual mission of *Water Resources Management*: advancing scholarship and guiding practice in managing water for a sustainable, equitable, and resilient future.

The following sections apply the structured framework to twenty global water challenges across irrigation, water supply, water quality, and water governance. Each challenge is analyzed in terms of its underlying issue, research priorities, and geographic relevance.

2 Irrigation

2.1 Water Conservation in Agriculture

2.1.1 Issue

Agricultural irrigation remains the world's largest user of freshwater, accounting for roughly 70% of global withdrawals. Yet much of this water is lost to evaporation, runoff, and deep percolation due to outdated infrastructure and inefficient application techniques (Fereres and Soriano 2007). These losses are especially problematic in arid and semi-arid regions, where growing urban and environmental water demands are intensifying competition with agriculture (Molden et al. 2010).

Ultimately, conserving agricultural water requires integrated responses: pricing reforms, legal frameworks that recognize and protect user rights, investment in efficient technologies, and support for collaborative governance structures. Without these changes, conservation will remain sporadic and insufficient to meet rising water demands in a changing climate.

2.1.2 Research Priorities

Research priorities include evaluating the cost-effectiveness of conservation incentives and understanding behavioral responses to different policy tools. Pilot programs in participatory irrigation management, remote sensing for water use verification, and real-time metering should also be scaled and studied in diverse agroecological settings. A few brief passages follow illustrating the kind of productive research that can be conducted in connection with finding innovative methods for promoting water conservation in irrigated agriculture through innovative water marketing arrangements. It is based on ongoing research work being conducted for North America's Rio Grande Basin in New Mexico (Fig. 1).

2.1.3 Example

Study Area To illustrate the potential of market-based institutional reforms for promoting agricultural water conservation, this paper presents an optimization modeling study from New Mexico's Rio Grande Basin, focused on two irrigation districts: Elephant Butte Irrigation District (EBID) (Elephant Butte Irrigation District, 2024) and Middle Rio Grande Conservancy District (MRGCD) (Middle Rio Grande Conservancy District, 2024) (Fig. 1). These districts differ in climate, water reliability, and cropping patterns, providing a valuable contrast. Both rely on limited surface water and compete for flows from the same river system (Booker et al. 2012). For the remaining 19 water issues, this paper reviews the literature, to keep the paper to a manageable length.

Data Observed data from 2017 to 2021 show that EBID averaged about 80,000 acres in irrigated production across eight major crops, while MRGCD averaged about 23,000 acres (Table 2). Irrigation water depth (Table 3) ranges from 2 to 5 feet annually, with EBID generally exhibiting higher ET due to hotter conditions in the Chihuahuan Desert.

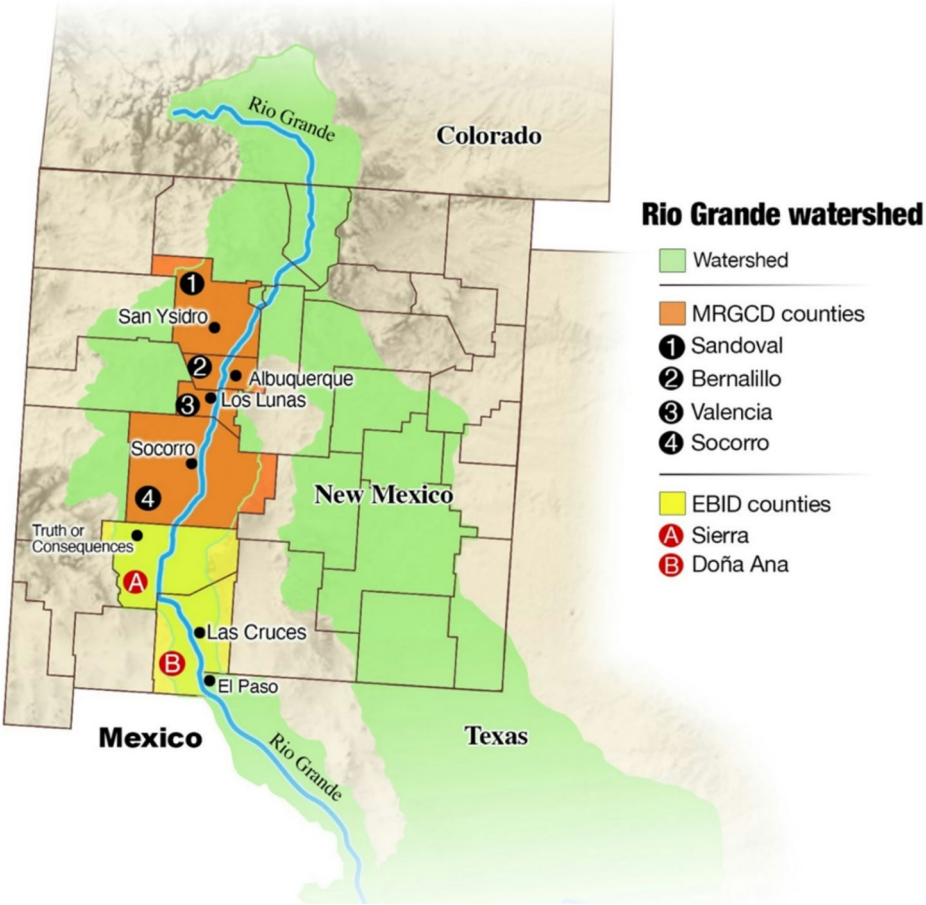

Fig. 1 New Mexico Rio Grande Basin Irrigation Districts, adapted from (Trail and Ward 2024)

Methods This model employed a positive mathematical programming (PMP) optimization model calibrated using observed crop choice, water use, and economic returns (Howitt 1995; Dagnino and Ward 2012; Trail and Ward 2024). The model assesses drought adaptation strategies under different water allocation institutions: proportional shortage sharing, intradistrict water trading, and interdistrict water trading.

Results Model results reveal that allowing interdistrict water trading significantly improves economic efficiency under drought compared to a proportional sharing of shortages. Table 4 shows optimized land use, where the highest percentage of historical acreage under drought occurs for the highest income earning crops, such as pecans and onions, while the lowest proportion of historic acreage under drought occurs for the lowest income crops, such as cotton and wheat. An identical proportion of historical acreage, 70%, occurs for all crops and districts when no trading of water for cash is permitted. Yield results (Table 5) demonstrate that with less land in production under shortage, yields improve slightly due to

Table 2 Aggregate measured cropland in production by crop, district, and year (acres/yr)

Crop	EBID					MRGCD				
	2017	2018	2019	2020	2021	2017	2018	2019	2020	2021
1_pecan	33000.6	33200.0	39215.6	41230.2	39989.5	141.2	147.6	1227.3	1325.6	1320.6
2_alfalfa	16827.5	17056.1	13098.8	14326.0	15396.4	19059.4	19919.2	15092.3	19107.9	21034.3
3_corn	5529.7	6174.2	4937.1	4443.3	3123.0	197.3	274.0	5008.5	3373.3	2160.9
4_wheat	3453.7	2587.7	428.9	3169.6	131.9	269.5	68.5	485.2	421.0	561.8
5_peppers	2924.4	3031.4	3212.5	2793.7	3350.0	18.1	37.9	1335.9	490.6	595.4
6_cotton	11112.2	14549.1	11873.3	10157.4	10804.1	25.3	98.1	3.2	36.0	68.4
7_onions	5490.4	3980.8	4266.0	3736.3	3564.9	23.3	56.2	8.9	105.3	43.4
8_beans	64.3	41.8	122.2	60.2	0.2	4.3	577.1	0.2	1.3	1.0

Table 3 Measured depth of water by crop, district, and year (Feet)

	EBID					MRGCD				
	2017	2018	2019	2020	2021	2017	2018	2019	2020	2021
1_pecan	3.91	3.72	3.61	3.91	3.49	3.48	2.76	4.32	4.01	2.40
2_alfalfa	3.73	3.55	3.45	3.74	3.33	4.00	3.64	3.26	3.14	3.04
3_corn	2.72	3.04	1.84	2.93	2.59	2.75	2.74	2.87	3.48	3.01
4_wheat	3.48	3.27	3.03	3.56	3.01	4.13	3.02	4.88	4.58	2.66
5_peppers	4.11	3.59	3.61	4.27	3.29	4.41	3.68	3.42	3.59	3.01
6_cotton	2.61	1.93	1.94	2.66	2.04	3.46	2.59	2.96	2.92	3.41
7_onions	2.65	3.47	2.30	3.32	2.71	2.39	3.60	3.41	2.89	3.56
8_beans	2.61	2.78	2.80	2.62	2.94	2.81	2.85	2.65	2.20	2.69

intensified use of higher-quality land, consistent with model calibration logic described in mathematical detail elsewhere (Trail and Ward 2024).

Optimized economic gains from water trading are summarized in Table 6. Interdistrict trading enables EBID to purchase water from MRGCD, increasing farm income in EBID while providing cash compensation to MRGCD. For example, MRGCD sells 11,171 acre-feet to EBID in exchange for $614,678 million in cash when interdistrict trading occurs. Overall, both districts benefit compared to proportional sharing, for which EBID gains $3,082,452 averaged over the five years and MRGCD gains $51,289 for the same period and trading arrangement.

Table 7 reports marginal values of water, revealing large differences between districts under alternative institutions. In 2017, the shadow price of water is $58.14 per acre foot in EBID versus $24.84 in MRGCD under intradistrict trading. Pecans, peppers, and onions show the highest shadow values under proportional shortage sharing, indicating their highest measured income per unit of water applied.

While this section describes a hydroeconomic model (Ward 2024) developed for the Rio Grande Basin, that basin is only illustrative. Where data are available or can be developed,

Table 4 Optimized cropland in production by district, crop, supply scenario, and water trading measure, averaged over all years, (Acres and proportion of Full)

Year	District	Crop	Water Supply Scenario						
			full	drought					
			Water Trading Institution						
			All	Inter district	Proportion full	Intra district	Proportion full	No trade	Proportion full
averaged	EBID	1_pecan	37327.2	32761.3	0.878	31726.6	0.850	26129.0	0.70
averaged	EBID	2_alfalfa	15341.0	9461.8	0.617	8009.6	0.522	10738.7	0.70
averaged	EBID	3_corn	4841.5	2818.8	0.582	2309.0	0.477	3389.0	0.70
averaged	EBID	4_wheat	1954.4	0.0	0.000	0.0	0.000	1368.1	0.70
averaged	EBID	5_peppers	3062.4	2828.5	0.924	2771.3	0.905	2143.7	0.70
averaged	EBID	6_cotton	11699.2	5431.1	0.464	3925.1	0.335	8189.5	0.70
averaged	EBID	7_onions	4207.7	4111.2	0.977	4087.1	0.971	2945.4	0.70
averaged	EBID	8_beans	57.7	35.2	0.609	29.4	0.509	40.4	0.70
averaged	MRGCD	1_pecan	832.5	677.1	0.813	733.4	0.881	582.7	0.70
averaged	MRGCD	2_alfalfa	18842.6	10057.3	0.534	13445.8	0.714	13189.8	0.70
averaged	MRGCD	3_corn	2202.8	747.0	0.339	1317.8	0.598	1541.9	0.70
averaged	MRGCD	4_wheat	361.2	0.0	0.000	0.0	0.000	252.8	0.70
averaged	MRGCD	5_peppers	495.6	427.3	0.862	454.8	0.918	346.9	0.70
averaged	MRGCD	6_cotton	46.2	7.2	0.156	18.7	0.405	32.3	0.70
averaged	MRGCD	7_onions	47.4	45.4	0.957	46.1	0.972	33.2	0.70
averaged	MRGCD	8_beans	116.8	37.7	0.323	66.6	0.570	81.7	0.70

Table 5 Optimized crop yield by district, crop, water availability scenario, and shortage sharing method, averaged over years (Tons per Acre)

Year	District	Crop	Water Supply Scenario			
			full	drought		
			Water Trading Institution			
			All	Inter district	Intra district	No trading
averaged	EBID	1_pecan	0.92	0.94	0.95	0.98
averaged	EBID	2_alfalfa	4.78	5.24	5.35	5.14
averaged	EBID	3_corn	4.06	4.47	4.57	4.37
averaged	EBID	4_wheat	0.72	0.90	0.90	0.78
averaged	EBID	5_peppers	7.54	7.68	7.72	8.11
averaged	EBID	6_cotton	0.47	0.53	0.54	0.50
averaged	EBID	7_onions	31.58	31.76	31.80	33.95
averaged	EBID	8_beans	1.05	1.15	1.18	1.13
averaged	MRGCD	1_pecan	0.64	0.67	0.66	0.69
averaged	MRGCD	2_alfalfa	3.78	4.22	4.05	4.06
averaged	MRGCD	3_corn	3.21	3.68	3.50	3.45
averaged	MRGCD	4_wheat	0.57	0.71	0.71	0.61
averaged	MRGCD	5_peppers	4.22	4.36	4.31	4.54
averaged	MRGCD	6_cotton	0.37	0.45	0.42	0.40
averaged	MRGCD	7_onions	17.68	17.88	17.81	19.01
averaged	MRGCD	8_beans	0.73	0.84	0.80	0.79

Table 6 Optimized economic gains from water trading arrangements for adapting to shortages relative to proportional shortage sharing, by crop, district, scope of trading, and hydrologic and economic indicator, averaged over years ($US)

Crop	Irrigation district	Inter District Trading				Intra District Trading			
		Water use change compared to proportional sharing of shortages (ac-ft/year averaged)	Gross income change compared to proportional sharing of shortages ($US/year averaged)	Economic value of water traded: + = bought water; - = sold water ($ US/year averaged)	Net income change compared to proportional sharing of shortages ($US/year averaged)	Water use change compared to proportional sharing of shortages (ac-ft/year averaged)	Gross income change compared to proportional sharing of shortages ($US/yr averaged)	Economic value of water traded: + = bought water; - = sold water ($US/yr averaged)	Net income change compared to proportional sharing of shortages ($US/yr averaged)
1_pecan	EBID	21,688	2,591,469	1,119,444	1,472,025	18,439	2,336,313	1,181,221	1,155,092
	MRGCD	269	27,616	14,633	12,984	449	37,949	14,916	23,034
2_alfalfa	EBID	−3,584	−234,375	−190,647	−43,729	−7,942	−566,041	−526,260	−39,781
	MRGCD	−8,958	−502,662	−465,241	−37,421	930	31,340	29,544	1,796
3_corn	EBID	−1,228	−69,190	−61,577	−7,612	−2,336	−150,381	−148,608	−1,773
	MRGCD	−2,025	−106,515	−110,925	4,410	−549	−21,696	−17,899	−3,798
4_wheat	EBID	−5,228	−51,181	−249,246	198,064	−5,228	−51,181	−305,968	254,786
	MRGCD	−1,110	−10,528	−58,363	47,835	−1,110	−10,528	−35,715	25,187
5_peppers	EBID	2,247	367,665	115,786	251,880	2,067	353,674	132,158	221,516
	MRGCD	238	27,102	13,042	14,061	317	31,737	10,458	21,279
6_cotton	EBID	−5,048	−275,337	−266,381	−8,956	−7,813	−488,399	−511,156	22,757
	MRGCD	−73	−2,745	−4,002	1,257	−36	−1,230	−1,242	12
7_onions	EBID	2,896	1,368,809	147,926	1,220,883	2,839	1,364,499	180,313	1,184,186
	MRGCD	33	9,319	1,691	7,628	35	9,433	1,177	8,255
8_beans	EBID	−12	−730	−627	−103	−25	−1,753	−1,700	−53
	MRGCD	−105	−4,978	−5,512	534	−37	−1,279	−1,239	−41
Total	EBID	11,731	3,697,130	614,678	3,082,452	0	2,796,731	0	2,796,731
	MRGCD	−11,731	−563,389	−614,678	51,289	0	75,726	0	75,726
Total	Both	0	3,133,741	0	3,133,741	0	2,872,457	0	2,872,457

Table 7 Optimized monetary economic value of additional water by year, crop, supply scenario, water shortage sharing institution, and irrigation district ($/acre-ft)

Year	Crop	Water Supply Scenario							
		Full	Drought						
		Water Shortage Sharing Method							
		All methods		Inter district		Intra district		No trading	
		District							
		EBID	MRGCD	EBID	MRGCD	EBID	MRGCD	EBID	MRGCD
2017	1_pecan	0	0	43.88	43.88	58.14	24.84	187.57	147.35
	2_alfalfa	0	0	43.88	43.88	58.14	24.84	34.75	25.57
	3_corn	0	0	43.88	43.88	58.14	24.84	27.60	21.62
	4_wheat	0	0	43.88	43.88	58.14	24.84	5.21	3.47
	5_peppers	0	0	43.88	43.88	58.14	24.84	205.29	107.08
	6_cotton	0	0	43.88	43.88	58.14	24.84	31.94	19.02
	7_onions	0	0	43.88	43.88	58.14	24.84	646.09	401.59
	8_beans	0	0	43.88	43.88	58.14	24.84	40.08	26.13
2018	1_pecan	0	0	52.51	52.51	64.32	33.35	138.75	130.74
	2_alfalfa	0	0	52.51	52.51	64.32	33.35	43.76	33.73
	3_corn	0	0	52.51	52.51	64.32	33.35	35.88	31.41
	4_wheat	0	0	52.51	52.51	64.32	33.35	3.03	2.59
	5_peppers	0	0	52.51	52.51	64.32	33.35	260.15	142.13
	6_cotton	0	0	52.51	52.51	64.32	33.35	35.32	20.76
	7_onions	0	0	52.51	52.51	64.32	33.35	442.56	239.01
	8_beans	0	0	52.51	52.51	64.32	33.35	33.98	23.21
2019	1_pecan	0	0	57.00	57.00	74.11	31.56	133.26	77.93
	2_alfalfa	0	0	57.00	57.00	74.11	31.56	42.88	35.79
	3_corn	0	0	57.00	57.00	74.11	31.56	42.19	21.36
	4_wheat	0	0	57.00	57.00	74.11	31.56	5.67	2.78
	5_peppers	0	0	57.00	57.00	74.11	31.56	207.98	123.06
	6_cotton	0	0	57.00	57.00	74.11	31.56	28.75	14.82
	7_onions	0	0	57.00	57.00	74.11	31.56	882.33	333.72
	8_beans	0	0	57.00	57.00	74.11	31.56	41.59	30.76
2020	1_pecan	0	0	46.59	46.59	52.22	34.13	87.25	59.63
	2_alfalfa	0	0	46.59	46.59	52.22	34.13	39.34	36.94
	3_corn	0	0	46.59	46.59	52.22	34.13	38.86	25.85
	4_wheat	0	0	46.59	46.59	52.22	34.13	5.07	3.12
	5_peppers	0	0	46.59	46.59	52.22	34.13	178.25	118.65
	6_cotton	0	0	46.59	46.59	52.22	34.13	22.28	16.04
	7_onions	0	0	46.59	46.59	52.22	34.13	722.52	465.65
	8_beans	0	0	46.59	46.59	52.22	34.13	38.14	31.73
2021	1_pecan	0	0	59.73	59.73	74.51	35.90	140.30	142.37
	2_alfalfa	0	0	59.73	59.73	74.51	35.90	42.01	36.34
	3_corn	0	0	59.73	59.73	74.51	35.90	47.98	32.63
	4_wheat	0	0	59.73	59.73	74.51	35.90	9.89	8.84
	5_peppers	0	0	59.73	59.73	74.51	35.90	190.43	116.78
	6_cotton	0	0	59.73	59.73	74.51	35.90	28.32	13.34
	7_onions	0	0	59.73	59.73	74.51	35.90	862.40	367.13
	8_beans	0	0	59.73	59.73	74.51	35.90	42.29	32.42

this same kind of analysis could be applied in many other parts of the world where there is competition for water among uses and economic sectors.

2.1.4 Geographic Scope of Problem

Water conservation in agriculture is a priority in many regions facing water scarcity, including the U.S. Southwest, northern India, Mexico's arid north, much of Iran, and southern Spain (Food and Agriculture Organization of the United Nations 2022).

2.2 Water Use in Agriculture

2.2.1 Issue

Inefficient water use in agriculture also contributes to declining water quality, as excessive runoff transports fertilizers, pesticides, and sediments into adjacent water bodies (Zhang et al. 2011). These diffuse pollution sources remain difficult to monitor and regulate, compounding environmental challenges and harming downstream communities and ecosystems.

Effective reform requires rethinking both supply and demand-side strategies. Investments in drip and sprinkler systems can substantially improve water productivity, but adoption remains limited in many regions due to financial, informational, and institutional barriers (Burney and Naylor 2012). Research shows that pairing water-saving technologies with supportive governance—such as tradable water rights or volumetric pricing—can significantly improve efficiency while maintaining crop yields (Gohar and Ward 2010).

2.2.2 Research Priorities

Future research could productively focus on quantifying the water savings and yield impacts of improved irrigation systems under diverse climate and soil conditions. Economic studies are needed to evaluate pricing reforms, subsidies, and trading mechanisms that encourage conservation. Integrated modeling efforts can also help evaluate how improved irrigation affects aquifer recharge, river flows, and long-term food security.

2.2.3 Geographical Extent of Problem

Regions where agricultural water use is particularly stressed include the Indo-Gangetic Plain, the North China Plain, California's Central Valley, the Colorado Basin, the Nile River Basin, and Australia's Murray–Darling Basin. In these and other areas, irrigation is often heavily subsidized, and efficiency gains are urgently needed to maintain long-term viability. One source showing the most affected regions globally is (Gassert 2013).

3 Water Supply

3.1 Climate Change and Water Challenges

3.1.1 Issue

The intensification of extreme weather events—droughts, floods, and storms—is further amplifying water-related risks for ecosystems, economies, and public policy (Eisenberg et al. 2007; Rogger et al. 2017; Abro et al. 2021). Irrigated agriculture is increasingly vulnerable to declining snowpack, shifting seasonal runoff, and heat-induced crop water loss (Cosgrove and Loucks 2015; Scanlon et al. 2017). Water supply reservoirs and aquifers are being drawn down more rapidly than they can recharge, especially in regions dependent on glacial melt or snowmelt for base flows (Rodell et al. 2009; Wada et al. 2012).

Altogether, climate change is transforming water from a renewable and locally managed resource into a globally interconnected risk multiplier. Lacking policy to promote adaptive planning and cross-sectoral coordination, climate-driven water shocks may destabilize food systems, displace populations, and intensify geopolitical tensions over shared water resources.

3.1.2 Research Priorities

Research needs to improve understanding of how climate change affects water supply reliability at the basin scale. Integrated climate-hydrology-economic models are needed to quantify risk and support adaptive management decisions. Such models should incorporate behavioral responses, cross-sector linkages, and institutional constraints. Further studies on probabilistic planning and real options analysis can aid investment under uncertainty.

3.1.3 Geographical Scope of Problem

Climate-related disruptions to water supply are evident across the globe. Western North America is experiencing record-low snowpacks and reduced reservoir inflows. Sub-Saharan Africa is seeing shifts in rainy season timing, affecting both municipal supplies and agricultural withdrawals. Southern Europe and parts of the Middle East face more frequent multi-year droughts, making it difficult to implement cross-border water agreements. Meanwhile, small island nations are confronting existential threats due to declining freshwater lenses from sea-level rise (Cochran 2021).

3.2 Climate Change and the Water Cycle

3.2.1 Issue

Observations from both satellite and ground-based monitoring confirm that snowmelt-dependent basins are especially vulnerable to earlier runoff, lower spring flows, and declining summer water availability (Barnett et al. 2005; Musselman et al. 2017). In parallel, intensifying drought frequency and severity have been observed across multiple continents,

exacerbated by rising evapotranspiration and land-atmosphere feedbacks (Cook et al. 2015; Diffenbaugh et al. 2015).

Aquifers and groundwater systems are under increased pressure as surface water becomes less reliable, driving greater reliance on groundwater pumping. However, recharge rates are also being affected by climate-induced shifts in precipitation timing and soil moisture dynamics (Cuthbert et al. 2019). At the same time, runoff intensity and variability are rising, leading to erosion, flash flooding, and lower infiltration—reducing the ability of landscapes to buffer water supply changes (Huntington 2006; Wasko and Sharma 2017). As climate signals strengthen, updating water infrastructure and management models to incorporate dynamic hydrological responses is no longer optional—it is essential.

3.2.2 Research Priorities

Research priorities include improved downscaling of climate projections to hydrologically meaningful spatial and temporal scales. Basin-level modeling should consider multiple climate futures and their implications for both surface and groundwater systems. The development of climate-smart decision tools, as well as better methods to incorporate uncertainty into operational and planning models, is needed. Comparative evaluations of adaptation strategies can inform policy choices tailored to local institutional and ecological contexts.

3.2.3 Geographic Scope of Problem

Himalayan and Andean river basins, snow-fed watersheds in North America, and glacial melt-dependent systems are particularly vulnerable to changes in snow and ice dynamics. Delta regions and coastal aquifers—such as those in Bangladesh, Vietnam, and the Nile Delta—face compounded risks from both hydrologic shifts and sea level rise (Kuzma et al. 2023).

3.3 Water Supply Unreliability

3.3.1 Issue

Unreliable water supply—manifested as seasonal interruptions, low-pressure events, or system failures—poses a critical threat to economic productivity, public health, and social equity worldwide. Water supply reliability is influenced by climatic variability, aging infrastructure, poor system design, and weak institutional capacity (Hashimoto et al. 1982; Sadoff et al. 2015). In both urban and rural settings, uncertainty in water availability undermines planning, deters investment, and drives reliance on unsafe alternatives.

Climate change further exacerbates unreliability through increased frequency of droughts, heatwaves, and altered precipitation patterns. These changes have already reduced the dependability of surface storage systems and stressed overdrawn aquifers (Milly et al. 2008; Rodell et al. 2009; Mekonnen and Hoekstra 2016; Zhu et al. 2021; Akpovi et al. 2022). Compounding the problem, many utilities in low- and middle-income countries experience systemic underfunding, resulting in deferred maintenance and inadequate service (McDonald et al. 2011; Kumpel et al. 2018).

Even in high-income countries, events like droughts in California or reservoir crises in Australia have highlighted the fragility of urban and agricultural supply systems under climate stress (Grafton et al. 2013). In all contexts, unreliability reflects both physical water stress and institutional failures in planning, pricing, and infrastructure management.

3.3.2 Research Priorities

Research should identify which supply configurations offer the best reliability for cost across varying contexts. Studies of social vulnerability to supply interruptions can inform equitable resilience planning. Hydraulic modeling of system interconnectivity and simulations of coordinated reservoir operations are valuable tools for managing complex water networks.

3.3.3 Geographic Scope of Problem

California, southern Africa, parts of the Middle East, and northeast Brazil all struggle with unreliability due to institutional, infrastructural, or climatic factors. In these areas, supply interruptions disproportionately affect agriculture and marginalized urban communities (World Bank Group 2020).

3.4 Water Supply Shortages

3.4.1 Issue

Drivers of chronic shortages are both natural and human-induced. Rapid urbanization, agricultural intensification, and industrial expansion have pushed many water systems beyond their sustainable limits, particularly in arid and semi-arid (Falkenmark and Rockström 2006; Vorösmarty et al. 2010). Climate change compounds these pressures, reducing snowpack, shifting precipitation patterns, and intensifying droughts that strain already over-allocated supplies (Milly et al. 2008; Diffenbaugh et al. 2015).

Shortages are often worsened by weak institutions that fail to adapt allocation rules to changing hydrologic realities. Water rights frameworks that allocate based on historical use rather than current availability contribute to overuse and inflexibility (Grafton et al. 2011). Additionally, limited investment in infrastructure, groundwater overextraction, and poor demand-side management contribute to the exhaustion of both surface and subsurface reserves (Rodell et al. 2009; Scanlon et al. 2012).

These supply deficits impose severe costs. Farmers abandon crops, cities impose rationing, and ecosystems collapse under reduced environmental flows. In many contexts, competition over scarce water has fueled legal disputes and interregional tensions (Petersen-Perlman et al. 2017). Addressing water shortages requires long-term structural reform—including improved pricing, allocation institutions, and investment in efficiency and storage—not merely crisis response.

3.4.2 Research Priorities

Future research should compare the lifecycle costs and resilience of alternative supply strategies. Studies of virtual water trade and the water footprint of agricultural exports are also essential. Socioeconomic research is needed to understand how communities adapt to chronic shortages and what support structures enable effective long-term adjustment.

3.4.3 Geographic Scope of Problem

Major cities like Cape Town, Chennai, and Mexico City, as well as entire basins like the Jordan and Indus, are experiencing structural deficits (Kuzma et al. 2023).

3.5 Aging Infrastructure

3.5.1 Issue

Aging water infrastructure leads to high water losses, contamination risks, and unaffordable maintenance burdens. The consequences of deferred infrastructure investment are both immediate and long-term. Chronic underfunding leads to backlogs in repair and replacement, resulting in water losses of up to 40% in some systems and significantly increasing the risk of service interruptions during climate extremes (Kleiner and Rajani 2001, 2002). Inadequate infrastructure also raises the likelihood of cross-contamination between drinking water and wastewater systems, particularly in areas with aging or combined sewer systems (Levin et al. 2002; Sadiq et al. 2004).

These vulnerabilities are further amplified by climate variability, population growth, and changing water quality requirements. More frequent droughts and floods stress pipes, storage reservoirs, and pumping stations not designed for such extremes (Milly et al. 2008). In developing regions, where infrastructure is often incomplete or poorly maintained, the risks are magnified, exacerbating water insecurity and sanitation challenges (Montgomery and Elimelech 2007).

Addressing the global infrastructure deficit demands coordinated investment, improved performance monitoring, and institutional capacity-building to maintain and modernize water systems for future demands.

3.5.2 Research Priorities

Cost-effectiveness studies of renewal schedules and upgrade packages can support financing plans. Research on governance models that ensure transparency and accountability in infrastructure spending is also needed. Addressing the challenge of aging water infrastructure requires interdisciplinary research that integrates engineering, economics, and policy analysis. In addition, studies are needed to evaluate financing mechanisms, including public-private partnerships and performance-based contracts, that can mobilize resources for large-scale upgrades. Finally, institutional research should examine governance structures and regulatory incentives that encourage proactive investment in infrastructure resilience, especially in low-income and rapidly urbanizing areas.

3.5.3 Geographic Scope of Problem

Cities with century-old water systems—e.g., New York, London, and Mumbai—face mounting pressure to modernize. Rapidly growing cities in Africa and Asia face parallel problems of overextension and insufficient capacity (UN Global Compact 2022).

3.6 Groundwater Depletion

3.6.1 Issue

In many agricultural regions, groundwater has enabled the expansion of irrigated farming beyond surface water limits. However, this development has often occurred without institutional safeguards or monitoring systems, leading to aquifer declines that are difficult to reverse (Konikow and Kendy 2005; Scanlon et al. 2012). In parts of India, China, Mexico, and the western United States, groundwater tables have fallen by tens of meters, increasing energy costs and salinization risks (Foster and Chilton 2003; Famiglietti 2014).

Policy responses remain uneven. In many countries, groundwater use remains largely unregulated or poorly enforced due to the difficulty of monitoring subsurface withdrawals. Even where institutions exist, political resistance to pumping restrictions impedes reform. Lacking stronger policy and governance, technological interventions alone—such as recharge basins or crop deficit irrigation—are unlikely to reverse depletion trends. Averting long-term collapse of groundwater resources requires integrated approaches that combine demand reduction, recharge enhancement, robust data systems, and enforceable legal frameworks.

3.6.2 Research Priorities

Interdisciplinary research on aquifer behavior, user incentives, and institutional evolution is needed. Models should incorporate both biophysical and socio-political dynamics. Data-sharing platforms and local monitoring can improve transparency and enforcement.

3.6.3 Geographic Scope of Problem

The Indo-Gangetic plain, Iran, central Mexico, the High Plains Aquifer in the U.S., and northern China are seeing critical depletion, with long-term consequences for agriculture and drinking water access. One GRACE map shows areas experiencing pumping and recharge (Jet Propulsion Laboratory, 2023).

3.7 Flooding and Stormwater Management

3.7.1 Issue

Urban development increases the volume and velocity of stormwater by replacing permeable surfaces with concrete, roofs, and roads, reducing natural infiltration and amplifying peak flows (Jacobson 2011). In many cities, drainage networks are poorly maintained or undersized to handle current rainfall intensities, let alone future extremes projected under

climate change (Güneralp et al. 2015). Combined sewer systems—still prevalent in older urban cores—can overflow during storms, discharging untreated sewage into rivers and homes (Roy et al. 2008).

Climate change further magnifies the problem, as warmer temperatures increase the intensity of rainfall events and extend wet-season extremes (Trenberth 2011; Prein et al. 2017). Even regions with declining annual precipitation can experience more destructive floods due to short-duration downpours that exceed infrastructure capacity. Inland and coastal floods have both increased in severity, often affecting the same areas already vulnerable to groundwater contamination, disease outbreaks, and economic instability (Douglas et al. 2008).

In low-income areas, stormwater mismanagement leads to stagnant water, vector-borne diseases, and recurring property damage. The urban poor often inhabit flood-prone lands where public infrastructure is limited or absent, exacerbating vulnerability (Douglas et al. 2008). Yet even affluent cities now experience regular "100-year floods" due to a combination of legacy infrastructure and extreme events.

Managing flood risk requires integrated watershed-scale strategies including green infrastructure, early warning systems, land use planning, and resilient drainage design. Traditional "grey" infrastructure must be complemented by decentralized and nature-based solutions.

3.7.2 Research Priorities

Effectiveness assessments of nature-based solutions, community-based drainage maintenance, and stormwater reuse are vital. Modeling runoff in dense, informal settlements can guide investments that reduce flood risk while improving public health. Research plans to address stormwater management and flooding at a global level should focus on both technological innovation and systems-level integration across urban, agricultural, and natural landscapes. Key research areas include: Together, these research initiatives can inform global adaptation strategies, support sustainable urban planning, and help reduce the escalating risks of urban flooding and water pollution.

3.7.3 Geographical Extent of Problem

High-risk cities include Jakarta, Lagos, New Orleans, and São Paulo. Informal settlements in these areas often face the brunt of flood-related losses due to poor drainage and limited services. For a visual representation of global flood risks (World Resources Institute 2015).

4 Water Quality

4.1 Wastewater Treatment and Reuse

4.1.1 Issue

Rapid urbanization, population growth, and inadequate infrastructure have led to widespread discharge of untreated or partially treated wastewater into rivers, lakes, and coastal waters.

In addition to biological waste, industrial pollutants such as heavy metals, pharmaceuticals, and chemicals from mining and manufacturing processes are increasingly entering aquatic systems. These pollutants can bioaccumulate, disrupt endocrine systems in wildlife and humans, and compromise agricultural productivity where contaminated water is used for irrigation (Miller 2006; Qadir et al. 2010; Angelakis and Snyder 2015; Jones et al. 2021; Breitenmoser et al. 2022; Finnerty et al. 2023).

Water pollution remains one of the most persistent and widespread water challenges internationally, undermining ecosystem health, economic productivity, and public well-being. Major pollutants include nutrients, pathogens, industrial chemicals, pharmaceuticals, microplastics, and heavy metals. These pollutants originate from a diverse array of sources, including untreated domestic sewage, agricultural runoff, mining activities, and industrial discharges. In many low- and middle-income countries, wastewater treatment coverage is incomplete or entirely absent, leading to direct contamination of rivers, lakes, and aquifers (Jones et al. 2021).

Urbanization and industrialization exacerbate pollution pressures. In rapidly growing cities, informal settlements often lack adequate sanitation infrastructure, leading to the discharge of raw sewage into nearby water bodies (Jones et al. 2021). In rural regions, excessive use of chemical inputs, coupled with inadequate soil management, accelerates leaching and surface runoff, transporting pollutants into aquifers and streams (Khatri and Tyagi 2015).

Additionally, global trade and manufacturing have led to the proliferation of microplastics in virtually all aquatic environments—from Arctic ice sheets to deep-ocean trenches—posing risks to aquatic organisms and potentially to human health via seafood consumption (Schwarzenbach et al. 2010a).

To address these multifaceted risks, scholars and policymakers have emphasized the importance of integrated water quality management approaches that span sectors, address both point and non-point sources, and employ tools such as pollution trading, ecosystem service valuation, and green infrastructure investments (Cude 2001; Rahman et al. 2009).

4.1.2 Research Priorities

More work is needed on cost-effective, energy-efficient treatment technologies that can operate in decentralized and off-grid settings. Membrane technologies, solar-powered filtration, and nature-based systems (e.g., wetlands) hold promise but require further validation. Policy research should evaluate the institutional frameworks that enable successful scaling of reuse systems, from water quality monitoring and certification to end-user acceptance and market incentives.

4.1.3 Geographical Extent

Major cities in South Asia (e.g., Dhaka, Karachi), Sub-Saharan Africa (e.g., Lagos, Nairobi), and Latin America (e.g., Lima, Mexico City) suffer from severe gaps in wastewater treatment. Even in high-income countries, aging sewer systems and combined stormwater overflows can lead to periodic pollution spikes, as seen in New York and London. Here is an international map characterizing the extent of the problem (UNU 2025).

4.2 Safe Drinking Water Access

4.2.1 Issue

Safe drinking water access is a fundamental human right and a prerequisite for public health, yet over two billion people still lack access to safely managed drinking water services worldwide (Sheel et al. 2024). The disparity is particularly stark in low-income and rural communities, informal urban settlements, and regions affected by conflict or displacement, where infrastructure gaps and affordability constraints converge (Cronk and Bartram 2017).

Unsafe drinking water contributes significantly to the global burden of disease, especially among children under five, through increased incidence of diarrhea, cholera, typhoid, and other waterborne illnesses. A global burden of disease analysis found that inadequate water and sanitation services accounted for hundreds of thousands of preventable deaths annually, mostly in low- and middle-income countries (Prüss-Ustün et al. 2019). Despite gains under the Millennium Development Goals, progress toward universal and equitable access remains uneven, especially when accounting for water quality and service continuity (Lee et al. 2012).

Even in cases where water sources exist, contamination from microbial pathogens, heavy metals, or industrial runoff often renders the water unsafe to drink. In some regions, natural contaminants such as arsenic or fluoride pose chronic health risks (Montgomery and Elimelech 2007). In others, aging and poorly maintained distribution systems lead to secondary contamination, particularly in unmonitored pipe networks (Simoes and Simões 2013).

Emerging technologies, such as household filtration systems, community-scale water treatment, and remote monitoring of microbial contamination, offer promising ways to expand access in underserved regions (Sobsey et al. 2008). Yet these innovations must be embedded in supportive institutional and policy frameworks to ensure sustainability and equity.

4.2.2 Research Priorities

Further studies are needed to test low-cost technologies in diverse environmental and cultural settings. Behavior changing interventions, gender-sensitive design, and supply chain logistics should be integrated into implementation research. Research on decentralized governance, blended finance, and mobile-based service delivery offers promising pathways for sustainably extending safe water access.

4.2.3 Geographical Extent of Problem

Sub-Saharan Africa, South Asia, and rural areas in Southeast Asia and Latin America face the most acute challenges in accessing microbiologically safe drinking water (World Health Organization 2025b).

4.3 Water Salinization

4.3.1 Issue

Water salinization poses a growing threat to agricultural productivity, freshwater availability, and ecosystem integrity, especially in arid and semi-arid regions. Salinity increases when soluble salts accumulate in water or soils, often as a result of poor drainage, high evaporation rates, or excessive groundwater pumping. In irrigated agriculture, the repeated application of water without sufficient leaching contributes to salt buildup, which impairs plant growth and reduces crop yields (Shrivastava and Kumar 2014).

In coastal regions, over-extraction of groundwater can induce seawater intrusion, contaminating aquifers that communities depend on for drinking and irrigation. This risk is accelerating as climate change raises sea levels and intensifies droughts, increasing reliance on groundwater and shrinking recharge opportunities (Ferguson and Gleeson 2012).

Ultimately, combating salinization requires integrated water and land management approaches tailored to local hydrogeologic and agronomic conditions. Multidisciplinary research and cross-sector collaboration are essential to develop scalable, cost-effective, and socially acceptable solutions (Tanji and Kielen 2002).

4.3.2 Research Priorities

Research on energy-efficient, small-scale desalination systems (e.g., reverse osmosis, capacitive deionization) is essential, particularly for rural areas. Salinity monitoring and early warning systems should be expanded. Socioeconomic research should examine the adoption barriers to salinity management practices and develop policy instruments that encourage uptake.

4.3.3 Geographical Extent of Problem

Salinity is a major concern in South Asia (e.g., Bangladesh, Gujarat), the Middle East, North Africa, and coastal China, as well as parts of Australia and the southwestern United States. A map can be found at (Food and Agriculture Organization of the United Nations 2025).

4.4 Water Safety

4.4.1 Issue

Water safety encompasses both the biological and chemical integrity of water intended for human use. While global access to water has expanded in recent decades, ensuring that water is actually safe to drink, bathe in, and cook with remains an urgent public health challenge. Unsafe water contributes to a significant global disease burden, particularly in low-income and rural settings (Prüss-Ustün et al. 2019). Even in regions with formal water supply infrastructure, inadequate treatment, intermittent supply, and contamination from distribution systems pose risks to consumers (Kumpel et al. 2018).

Chemical threats—such as arsenic, fluoride, and nitrates—occur both naturally and as a result of agricultural and industrial activities. Arsenic contamination in South Asia and fluo-

ride in East Africa are prominent examples of chronic, long-term health threats from naturally occurring substances in groundwater (Montgomery and Elimelech 2007). Meanwhile, nitrates from fertilizer runoff increasingly threaten rural groundwater sources in agricultural zones (Ward et al. 2005).

Water safety also includes perceptions and trust, which strongly influence behavior and policy. When users lack confidence in municipal water systems, they often turn to expensive and sometimes riskier alternatives such as bottled water, private tankers, or untreated sources (Shadabi and Ward 2022). Restoring trust in public systems requires not only technical reliability but also transparency, consistent service, and inclusive engagement with affected communities.

The WHO's Water Safety Plan (WSP) framework is increasingly seen as a gold standard for managing risks systematically from source to tap. However, implementation remains limited in many countries due to weak institutional capacity, lack of monitoring data, and fragmented responsibilities. Progress in water safety will require coordinated investment in infrastructure, capacity building, and behavior-focused public health interventions.

4.4.2 Research Priorities

Research should develop methods to detect, trace, and mitigate emerging contaminants. Multi-pollutant modeling and watershed-level analysis can improve regulatory design. Interdisciplinary studies that link water quality to economic productivity, biodiversity, and public health outcomes are also needed to inform integrated planning. Community-based monitoring systems can empower local stakeholders to track water quality and demand better management from authorities (Cude 2001; Santhi et al. 2006; Koncagül et al. 2018).

4.4.3 Geographic Scope of Problem

Major rivers such as the Ganges, Yangtze, Mississippi, and Danube are increasingly impacted by nutrient loads and industrial discharges. Rapidly industrializing countries in Asia and Latin America face growing water quality challenges alongside economic development. For a visual representation of global water safety, see the interactive maps at (UN and GEMStat 2025).

4.5 Water Pollution

4.5.1 Issue

Water pollution remains one of the most persistent and globally pervasive threats to human health, biodiversity, and water security. Pollutants including nutrients, pathogens, industrial effluents, pharmaceuticals, plastics, and heavy metals continue to degrade rivers, aquifers, and coastal ecosystems across both developed and developing nations (Schwarzenbach et al. 2010b). Rapid urbanization and industrial expansion, combined with weak enforcement and aging infrastructure, exacerbate the challenge, especially in low-income countries where wastewater treatment coverage is limited or absent (Jones et al. 2021).

Agricultural non-point source pollution is one of the most widespread and difficult-to-control contributors to global water quality degradation. Runoff from fertilizers and pesti-

cides promotes eutrophication, leading to hypoxic zones and biodiversity collapse in lakes, estuaries, and seas—including the Gulf of Mexico, the Baltic Sea, and parts of the Bay of Bengal (Postel et al. 1996). Meanwhile, pesticide residues and nitrate contamination of rural groundwater sources remain a significant human health concern (Khatri and Tyagi 2015).

Effective water quality management requires cross-sectoral coordination and integrated watershed approaches. While regulatory measures such as effluent standards and discharge permits have shown some success, they must be combined with incentives for pollution prevention and investment in green infrastructure, such as constructed wetlands and riparian buffers (Cude 2001). Community-based monitoring and education campaigns also play a critical role in raising awareness and improving compliance.

4.5.2 Research Priorities

Addressing water pollution necessitates a comprehensive research approach focusing on prevention, monitoring, and remediation, including:

- Source Identification and Control: Investigate the origins of pollutants to implement targeted reduction strategies. This includes studying industrial discharge, agricultural practices, and urban runoff. Research should aim to develop best management practices (BMPs) that minimize pollutant entry into water bodies.
- Advanced Monitoring Technologies: Develop and deploy sensors capable of real-time detection of contaminants, including emerging pollutants like microplastics and PFAS. This involves creating cost-effective, sensitive, and durable monitoring systems that can be widely implemented.
- Policy and Community Engagement: Study the effectiveness of regulatory frameworks and community-based initiatives in reducing water pollution. This includes assessing the impact of policies like the US Clean Water Act and exploring community-led watershed management programs.

4.5.3 Geographic Scope of Problem

For a visual representation of water pollution levels by country, see (GEMS Water Data Centre 2025).

4.6 Waterborne Diseases and Sanitation

4.6.1 Issue

Despite decades of investment in infrastructure and hygiene education, over 1.5 million deaths per year are still linked to diseases such as diarrhea, cholera, typhoid, and hepatitis A (Prüss-Ustün et al. 2019). These illnesses disproportionately affect children under five, with contaminated drinking water and inadequate sanitation among the primary drivers (Freeman et al. 2017).

In rural and peri-urban areas, open defecation, insufficient toilet coverage, and poor fecal sludge management contribute to the contamination of drinking water sources. Human waste often enters shallow aquifers and rivers used for domestic supply, especially during

rainy seasons when latrines overflow or collapse (Cabral 2010). Even where toilets exist, lack of handwashing infrastructure and behavior gaps continue to undermine disease control (Fink et al. 2011).

Effective prevention of waterborne disease requires integrated approaches that combine improved sanitation hardware, hygiene behavior change, and community-level engagement. School-based sanitation interventions, household handwashing campaigns, and women's water user associations have all demonstrated success when linked to robust institutional support and monitoring (Cronk and Bartram 2017). Sustained health gains will depend not only on infrastructure investments, but also on tackling inequality in service provision, prioritizing vulnerable groups, and addressing upstream water contamination sources.

4.6.2 Research Priorities

To further mitigate the impact of waterborne diseases and improve sanitation, targeted research is essential in the following areas:

- Disease Surveillance and Reporting: Enhancing the collection and analysis of data on waterborne disease outbreaks can inform public health interventions. For example, the US Centers for Disease Control and Prevention reported 214 disease outbreaks brought on by poor drinking water in the US between 2015 and 2020, underscoring the need for robust surveillance systems.
- Impact Evaluation of WASH Interventions: Assessing the effectiveness of various WASH programs can identify best practices and areas for improvement. Research into the sustainability and scalability of management interventions like the Sato pan can guide future investments.

4.6.3 Geographic Scope of Problem

For a visual representation of the global burden of waterborne diseases and sanitation, the World Health Organization provides data on the burden of disease attributable to unsafe WASH by country. This resource includes maps and statistics that highlight disparities in disease burden related to water and sanitation (World Health Organization 2025a).

5 Water Governance

5.1 Conflicts and Disputes Over Water

5.1.1 Issue

Conflicts over water are escalating worldwide as growing demand intersects with climate-induced scarcity, governance failures, and uneven institutional capacity. These disputes arise not only between countries but also within nations, across regions, sectors, and user groups. Water governance is often fragmented among overlapping agencies and jurisdictions, leading to policy incoherence and legal ambiguity (Tropp 2007; Abson et al. 2014; Cosgrove and Loucks 2015). In federal systems and rapidly urbanizing regions, conflicting

mandates and competition for control of scarce resources have become common flashpoints (Zeitoun and Mirumachi 2008; De Stefano et al. 2017).

Climate change compounds these tensions by reducing water predictability and increasing the frequency of extreme events. In drought-prone regions, intensified scarcity exacerbates existing social and political divisions, raising the risk of conflict (Dinar et al. 2015). Moreover, weak or absent dispute resolution mechanisms within water governance institutions often allow such tensions to escalate (Giordano and Wolf 2003).

Ultimately, improving dispute prevention and resolution in the water sector requires institutional transparency, equitable access frameworks, and mechanisms for multistakeholder participation. Policymakers and scholars alike emphasize the importance of adaptive, polycentric governance systems that can navigate complexity while promoting cooperation across sectors and scales.

5.1.2 Research Priorities

Identify legal and institutional models that support polycentric governance, resilience, and equitable access across different political and economic systems. Being able to identify those models will be a big help in promoting improved water governance.

5.1.3 Geographical Extent of Problem

Institutional fragmentation is prevalent in federal systems (e.g., India, the United States, Brazil) and in post-conflict or rapidly urbanizing regions lacking regulatory consistency. Here is a map showing current global water hotspots (WPS 2025).

5.2 Transboundary Water Conflicts

5.2.1 Issue

Transboundary water conflicts remain one of the most persistent and geopolitically destabilizing forms of environmental conflict in the 21st century. These disputes arise when multiple political entities share freshwater systems—rivers, lakes, or aquifers—but lack coordinated governance frameworks for their allocation and management. Fragmented jurisdictional authority, asymmetric bargaining power, and weak institutional arrangements often result in tension, especially where water is scarce or development projects (e.g., large dams, irrigation diversions) are perceived as threats by downstream nations (Wolf et al. 2003; Zeitoun and Warner 2006).

A growing body of research emphasizes that cooperation over shared waters is both possible and beneficial. However, such cooperation depends on long-term institution-building, mutual economic incentives, and regional political will (Sadoff and Grey 2002; Giordano and Wolf 2003). Thus, resolving transboundary water conflicts is as much about political economy and governance reform as it is about hydrology.

5.2.2 Research Priorities

Evaluate the effectiveness of anti-corruption strategies in water projects. Study how open governance improves compliance and allocation equity. Economically responsible actions to address transboundary water conflicts should prioritize cooperative management frameworks that balance equitable water access with efficient resource use. Countries sharing rivers, lakes, or aquifers can benefit from establishing joint water commissions, data-sharing agreements, and treaties that allocate water based on negotiated needs, seasonal variability, and ecological considerations. Economic tools such as benefit-sharing mechanisms—where upstream and downstream users collaborate on hydropower, irrigation, or conservation projects—can align incentives and reduce tensions.

5.2.3 Geographic Scope of Problem

Corruption in water governance is a challenge in many low- and middle-income countries, especially where discretionary power and opaque financing dominate decision-making. For a visual representation of transboundary water basins and associated conflicts, see (Oregon State and University 2025).

5.3 Water Inequality and Access

5.3.1 Issue

Persistent inequality in access to safe and affordable water services continues to pose a major public health and human rights challenge globally. Despite recognition of water as a basic human need, over two billion people still lack reliable access to safely managed drinking water (Sheel et al. 2024). These disparities are most severe in rural areas, low-income urban settlements, and Indigenous communities, where infrastructure gaps and financial constraints intersect to undermine service delivery (Cronk and Bartram 2017; Bray 2021).

The roots of water inequality are both structural and institutional. Utilities in many regions operate under financial deficits due to politically suppressed tariffs, leading to chronic underinvestment in infrastructure (Truelove 2011; Shadabi and Ward 2022). This results in patchy coverage, intermittent supply, and poor water quality—conditions that disproportionately affect vulnerable populations. In addition, informal settlements are often excluded from official planning frameworks, further exacerbating service gaps (Fink et al. 2011).

Ensuring equitable water access is not only a matter of social justice—it is also essential for public health, economic development, and climate adaptation. Closing the water access gap demands structural reforms in utility governance, infrastructure finance, and pricing policy that foreground the needs of marginalized users while ensuring system viability.

5.3.2 Research Priorities

Model the behavioral impacts of different pricing strategies. Study tariff reform impacts on vulnerable populations and utility sustainability. This kind of behavior modelling stands to make a great contribution to reducing water inequality and improving access.

5.3.3 Geographic Scope of Problem

Subsidized or flat-rate pricing persists in many agricultural regions (e.g., Middle East, Central Asia) and urban utilities across Africa, Latin America, and South Asia. For a visual representation of global water access disparities, the World Resources Institute provides an interactive map highlighting areas facing high water stress (National and Geographic 2023).

5.4 Ineffective Water Governance

5.4.1 Issue

Ineffective water governance remains one of the most formidable barriers to sustainable water management worldwide. Governance failures—ranging from fragmented institutional responsibilities to weak enforcement of regulations—undermine efforts to equitably allocate, efficiently use, and sustainably protect water resources (Pahl-Wostl et al. 2008). These failures are especially prevalent in regions with overlapping jurisdictions or unclear authority, where conflicting mandates lead to gridlock and policy incoherence (Tropp 2007; Lautze et al. 2011).

In federal systems, such as those in the United States, India, and Brazil, uncoordinated governance often results in inefficient water use and inter-jurisdictional conflict (Gupta and Pahl-Wostl 2013). Water governance is further weakened by limited stakeholder participation, lack of transparency, and insufficient integration of scientific knowledge into decision-making (Biswas and Tortajada 2010). The result is a policy landscape unable to adapt to emerging challenges such as climate change, urban growth, and agricultural intensification (Garrick et al. 2013).

Moreover, political instability and capacity constraints often leave water institutions underfunded and ill-equipped to enforce regulations, respond to crises, or implement adaptive policies. In the face of weak improvements in institutional design and governance capacity, technical solutions alone cannot deliver water security.

5.4.2 Research Priorities

Examine institutional designs that foster durable cooperation, especially under stress scenarios like drought or upstream development. Policy research should investigate the effectiveness of water pricing mechanisms, water rights allocation, and the contribution of international agreements in influencing the allocation of shared water supplies (Savenije and Van Der Zaag 2002).

5.4.3 Geographical Extent of Problem

Tensions persist in the Nile, Tigris-Euphrates, and Mekong basins. Even in North America and Europe, subnational disputes over water allocation are increasing. A visual representation of water governance levels by country can be found at a 2018 work (United Nations - UN Water 2018).

5.5 Water Governance and Corruption

5.5.1 Issue

Conflicts over water resources are intensifying globally, arising from competing demands among agricultural, urban, industrial, and environmental users. These disputes span scales—from local farmer-user conflicts to international tensions across shared river basins—and reflect structural mismatches between growing water demand and finite or declining supply (Zeitoun and Mirumachi 2008; De Stefano et al. 2017). The problem is often exacerbated by fragmented governance, ambiguous water rights, and uneven institutional capacity across jurisdictions.

Climate change is accelerating the likelihood of water disputes by altering hydrological patterns, amplifying scarcity, and increasing unpredictability in water (Hoekstra and Mekonnen 2012; Diffenbaugh et al. 2015; Elahi et al. 2024; Abro et al. 2025). In many parts of the world, especially arid and semi-arid regions, droughts are becoming more frequent and prolonged, increasing the stakes in water allocation decisions and hardening political positions (Cook et al. 2015). Even in high-income nations, declining snowpack, groundwater overdraft, and population growth are heightening tensions among users (Gerlak et al. 2012).

Effectively addressing water conflicts requires institutional clarity, transparent allocation rules, conflict resolution mechanisms, and inclusive stakeholder participation. Without reforms that promote equity, trust, and adaptability, disputes over water are likely to escalate in frequency and severity under future climate and development pressures.

5.5.2 Research Priorities

Study how data transparency influences behavior and governance outcomes. Develop institutional models that link data to decision-making in adaptive frameworks. Research plans to address water governance and corruption should focus on identifying institutional weaknesses, evaluating anti-corruption strategies, and designing governance frameworks that promote transparency, accountability, and efficiency. Quantitative analyses using governance indicators, financial flows, and service delivery outcomes can help detect patterns of corruption or inefficiency. Experimental and behavioral research—such as randomized controlled trials testing the impact of digital monitoring tools, community audits, or transparency campaigns—can reveal what interventions most effectively reduce corruption and improve trust. Interdisciplinary studies should also examine how legal frameworks, power dynamics, and social norms influence governance outcomes.

5.5.3 Geographic Scope of Problem

Data gaps are especially problematic in developing countries and remote regions, where institutions rely on sparse or outdated hydrological records. Here is an interactive map displaying the extent of perceived public sector corruption globally (Transparency International 2023).

6 Discussion: Cross-Cutting Insights for Policy and Practice

The twenty global water challenges reviewed in this paper reveal several powerful cross-cutting themes that transcend individual sectors and regions. Despite their diversity, these challenges share common institutional, economic, and hydrologic roots—suggesting that integrated, systems-based solutions are both necessary and achievable. Across irrigation, supply, quality, and governance domains, three recurring drivers stand out: chronic under-investment in infrastructure, fragmented institutional frameworks, and failure to price water in ways that reflect scarcity and promote conservation.

Tradeoffs are also inherent in water policy choices. Efforts to improve supply reliability may require large infrastructure investments that could exacerbate ecological degradation or inequity if poorly designed. Similarly, pricing reforms may improve efficiency but raise concerns over affordability and access. The most effective strategies recognize these tensions and build mechanisms—such as cross-subsidies or stakeholder negotiations—that reconcile competing goals. The reviewed literature and modeling evidence both point toward institutional design as the critical determinant of whether water management strategies succeed in balancing economic efficiency, environmental sustainability, and social equity.

In contributing a globally structured synthesis combined with original modeling results, this paper advances the mission of *Water Resources Management* by offering a practical framework for decision support under complexity. It highlights not only where the most urgent research and reform needs lie but also how integrated modeling and policy tools can inform adaptive responses. Future research should deepen this integration by building coupled hydrologic-economic models across diverse geographic settings, evaluating tradeoffs under uncertainty, and testing policy reforms through participatory, stakeholder-driven processes. The path forward lies not in choosing between science and governance, but in bridging them.

This review article's focus on 20 globally important water issues has included much, but the constraints of journal space require leaving much untouched. Much future work along the lines of this paper need to give serious attention to recent works on scholarship addressing machine learning in water forecasting, water justice, and circular water economy, equity across socio-economic classes, genders, and indigenous groups, all of which are largely untouched in this work.

7 Conclusions

This paper has synthesized twenty of the most pressing global water management challenges—spanning irrigation, water supply, water quality, and water governance—into a structured, integrated framework that supports decision-making in both research and policy (Table 8). Each challenge is examined in terms of its underlying issue, research priorities, and geographic relevance. Collectively, the framework reveals a shared set of vulnerabilities facing water systems worldwide: increasing climate variability, aging infrastructure, institutional fragmentation, and persistent inequities in access and quality.

What distinguishes this contribution is its combination of systematic synthesis with original empirical modeling. The hydro-economic analysis of interdistrict and intradistrict water trading in North America's Rio Grande Basin demonstrates that institutional reform can

Table 8 Summary of global water challenges

Challenge	Geographic Relevance	Key Research Gaps
1. Water Conservation in Agriculture	U.S. Southwest, northern India, Mexico, Iran, Spain	Behavioral and economic analysis of conservation incentives
2. Water Use in Agriculture	Indo-Gangetic Plain, North China Plain, California, Nile Basin, Australia	Economic studies on pricing and trading impacts
3. Climate Change and Water Challenges	Western U.S., Sub-Saharan Africa, South Asia, Mediterranean, Small Islands	Climate-hydrology-economic modeling under uncertainty
4. Climate Change and the Water Cycle	Himalayas, Andes, snow-fed basins, coastal aquifers in Asia and Africa	Downscaled climate and hydrology impact modeling
5. Water Supply Unreliability	California, southern Africa, Middle East, northeast Brazil	Design of reliable, cost-effective supply configurations
6. Water Supply Shortages	Cape Town, Chennai, Mexico City, Jordan Basin, Indus Basin	Lifecycle costs and socio-adaptation under shortages
7. Aging Infrastructure	New York, London, Mumbai, African and Asian cities	Cost-effective investment and governance performance
8. Groundwater Depletion	India, China, Mexico, U.S. High Plains, northern China	Integrated socio-hydrological modeling and regulation
9. Flooding and Stormwater Management	Jakarta, Lagos, New Orleans, São Paulo	Effectiveness of green infrastructure and forecasting tools
10. Wastewater Treatment and Reuse	Dhaka, Karachi, Lagos, Nairobi, Lima, NYC, London	Energy-efficient reuse tech and institutional scaling
11. Safe Drinking Water Access	Sub-Saharan Africa, South Asia, Southeast Asia, Latin America	Behavioral, technical, and decentralized governance models
12. Water Salinization	Bangladesh, Gujarat, Middle East, North Africa, Australia, U.S. Southwest	Desalination, salinity monitoring, and farmer incentives
13. Water Safety	Ganges, Yangtze, Mississippi, Danube, industrial Asia and Latin America	Multi-pollutant risk modeling and real-time monitoring
14. Water Pollution	Global (esp. Gulf of Mexico, Baltic Sea, Bay of Bengal)	Source tracking, advanced sensors, and institutional incentives
15. Waterborne Diseases and Sanitation	Sub-Saharan Africa, South Asia, Latin America, global slums	Surveillance, WASH impact evaluation, sanitation behavior
16. Conflicts and Disputes over Water	India, U.S., Brazil, post-conflict regions	Polycentric governance and conflict resolution frameworks
17. Transboundary Water Conflicts	Nile, Indus, Mekong, Tigris-Euphrates	Benefit-sharing, data-sharing, regional collaboration tools
18. Water Inequality and Access	Middle East, Central Asia, Africa, South Asia, Latin America	Pricing impacts on vulnerable users and access strategies
19. Ineffective Water Governance	U.S., India, Brazil, Europe, Mekong and Nile Basins	Institutional design for coordination and integration
20. Water Governance and Corruption	Global—esp. low- and middle-income countries	Transparency tools, anti-corruption strategies, RCTs

significantly enhance economic efficiency, resource-use equity, and resilience for handling climate-induced drought stress. This real-world example reinforces the broader insight that institutional design—not just physical infrastructure—is central to meeting emerging water challenges.

Looking ahead, future research should deepen the integration of hydrologic, economic, and institutional models; expand the use of participatory scenario planning; and test innovations in real-world settings where risk, uncertainty, and stakeholder diversity are greatest. Researchers, practitioners, and policymakers alike need to build bridges across disciplines to co-develop solutions that are scientifically sound, economically viable, and socially just.

In an era of rising water stress, this paper offers both a panoramic view of the challenges and a grounded roadmap for addressing them—through better science, smarter institutions, and more inclusive strategies for managing the world's most essential resource.

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Declarations

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